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APPLIED RESEARCH

SaaS API Monitoring for Data Leakage and Access Anomalies: A Privacy-Aware and Adaptive Behavioral Analytics Framework for Enterprise Cloud Systems

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ABSTRACT Cloud-based Software as a Service (SaaS) generates large volumes of activity using dynamic, API-driven services, and traditional tools used for metadata scanning are unable to keep up with real-time visibility into that activity. Enterprise solutions that rely upon periodic or incremental scanning for that activity (e.g., Microsoft Purview) are unable to detect changes in permissions, exposure of files and anomalous user behavior. In this paper, we propose a solution that provides the ability to monitor and enforce policies in real-time at the firewall level through the use of a man-in-the-middle (MITM) proxy that allows for round-trip blocking of either users or permissions based on suspicious activity identified through visibility into file operations and sharing between users. This solution is designed to operate in conjunction with Microsoft Presidio to detect personally identifiable information (PII), utilizes an exponentially weighted moving average (EWMA)-based behavioral model for adaptive anomaly detection, and perform multi-cloud monitoring of SaaS solutions. A single administrative dashboard aggregates all events, sensitivity scores, relationships between permissions, anomalies and automatically enforced actions, thereby providing end-to-end visibility, fine-grained traceability, rapid response to high-risk events, and extensive cross-platform coverage. The privacy detection module powered by Microsoft Presidio achieved over 96% accuracy in identifying PII, which lays the groundwork for providing real-time, privacy-aware multi-cloud SaaS security.

INDEX TERMS Cloud security, SaaS monitoring, cloud access security broker (CASB), API interception, PII detection, vector database, pinecone, Presidio, EWMA, anomaly detection.

I. INTRODUCTION

Enterprise environments are rapidly evolving in the way they store, access, and share information, driven by the widespread adoption of SaaS and multi-cloud ecosystems. The increasing reliance on link-based collaboration has significantly elevated the risk that even minor permission changes can

unintentionally expose sensitive information. Such exposure paths are often difficult to identify using traditional perimeter-based security controls [1]. These challenges highlight the limitations of legacy Data Loss Prevention (DLP) models, which primarily rely on rule-based scanning or log ingestion and lack the capability to provide real-time visibility into the propagation of permissions.

Recent advancements in metadata-driven DLP approaches have emphasized the importance of monitoring file-level

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attributes, sharing configurations, and contextual indicators as key security signals for preventing unintended exposure of cloud-stored data [2]. However, metadata alone is often insufficient to capture the true sensitivity of content, particularly as documents evolve over time through continuous edits and contributions from multiple users. Vector databases address semantic risk by enabling large-scale, high-dimensional similarity search over embedded representations of data [3]. By establishing relationships between semantically similar content, these systems allow for the identification of exposure patterns that may not be detectable through purely structural or rule-based methods. In parallel, established tools for detecting personally identifiable information (PII), such as Microsoft Presidio, provide reliable mechanisms for extracting sensitive entities from diverse file formats [4], enabling more context-aware and consistent identification within security workflows.

Despite these advancements, a key gap remains in integrating content sensitivity with permission dynamics. Historical incidents have shown that even trusted users can unintentionally expand access to sensitive data, resulting in large-scale exposure [1]. This highlights a persistent disconnect in existing systems, which often treat permission states and content sensitivity as independent factors rather than interdependent dimensions in real-world usage.

To address this gap, we propose an event-based monitoring architecture that combines semantic content analysis with real-time policy enforcement. The system leverages vector embeddings for capturing semantic relationships, integrates PII detection for sensitivity classification, and enables real-time interception of permission changes. Additionally, adaptive behavioral modeling using an EWMA-based anomaly detection approach is employed to establish dynamic baselines and identify unusual sharing patterns. By combining behavioral analytics with semantic intelligence, the proposed framework enables proactive detection and mitigation of both unintentional and malicious data exposure in SaaS environments.

II. LITERATURE REVIEW

Monitoring sensitive data access and detecting anomalous behavior in Software-as-a-Service (SaaS) platforms draws from three major research streams: cloud anomaly detection, PII detection with privacy-preserving processing and security governance in traffic analytics. This section reviews the most relevant contributions in both domains and highlights key limitations motivating this work.

A. ANOMALY DETECTION IN CLOUD AND NETWORK ENVIRONMENTS

A prominent area of research in the area of cloud security is anomaly detection, particularly for the purpose of detecting anomalies and monitoring large-scale API-centric systems. An emerging trend related to anomaly detection is the use of statistical techniques and machine learning for the purpose

of performing real-time monitoring of the cloud APIs. Whiteman et al. [1] conducted studies on the use of anomaly detection to detect abnormal behavior of the API traffic through the use of real-time monitoring, while demonstrating the advantages of using adaptive monitoring for improving operational visibility. In their study, Islamic et al. [2] described a dataset from large industrial companies that can be used for the purpose of diagnosing anomalies in the cloud and listed several challenges related to the development of high-dimensional telemetry in the cloud. Shahzad et al. [3] also have identified a use of a classification of anomaly detection using ensembles, and have attempted to build a mass record of the fact that many of the uses of laureate (last term) classification techniques can be implemented in the context of multiclass anomaly detection. The use of generative artificial intelligence-based systems has also been applied to the area of expert systems built for purpose of cloud anomaly detection, and these tools also have been shown to provide greater opportunities for discovering patterns using richer contextual elements [4]. Hagemann and Katsarou's [5] systematic review of the literature provides an overview of the various anomaly detection techniques in the cloud across a variety of cloud layers. Another example of a prominent technique for detecting anomalies in time-series data is the use of EWMA techniques (Hoblos [6]); An EWMA is proposed for use with univariate data, while Li and Tsikerdekis [7] have developed a method of using ERGM and ARMA to model network traffic in this area. Knowledge graph embeddings also support semantic-rich anomaly identification in distributed cloud systems [8]. Federated learning-based detection provides a means of privacy-preserving collaborative modeling in multi-tenant platforms, as in Wang et al. [9]. Most recently, there is active research in LLM-assisted anomaly detection to augment SRE workflows and enhance cloud reliability [10].

B. PII DETECTION, DATA SANITIZATION, AND PRIVACY PRESERVATION

The implementation of PII detection has grown to include cloud environments utilizing machine learning (ML), Natural Language Processing (NLP), and synthetic datasets to increase privacy. In particular, Mishra et al. [11] describe a PII extraction and anonymization method that combines rules-based and ML techniques with excellent results for accuracy with respect to entities sensitive to PII. Query-aware privacy-preservation tools such as PII-Bench [12] offer benchmarks for assessing the trustworthiness of language-based systems while also proposing GLINER-based obfuscation methods for lineage tracking to assist in the evaluation of language services as well as improving compliance with the PII definition via contextual classification (Shivaprakash et al. [13]). Synthetic dataset generation for PII detection is also growing with Savkin et al.'s [14] implementation of SPY. Efforts towards de-identifying larger data sets include the work of Larson et al. [15] who look to sanitize larger scale data sets

based on archival documents. Mainetti and Elia [16] have developed a standardization of ML-P-based PII detection that will ultimately help organizations meet their privacy obligations. Cloud-based DLP systems such as CloudDLP [17], and patent-pending metadata-based DLP systems developed by Google [24] offer the ability to perform non-disruptive sanitation of sensitive data. Microsoft Presidio [25], another widely utilized solution for deterministic recognition of PII, operates hand-in-hand with these other industry solutions for enterprise privacy. These combined efforts result in privacy-aware monitoring systems that can be used real-time to assess API traffic, content of files, and meta-data associated with operations.

C. SECURITY GOVERNANCE, TRAFFIC ANALYTICS, AND CLOUD ACCESS CONTROL

As the field of cloud security research evolves, more studies are incorporating anomaly detection techniques into traditional governance mechanisms such as network analytics, proxy-level inspections, data loss prevention (DLP), and SaaS application monitoring. One recent example of this convergence of anomaly detection and traditional governance mechanisms can be found in a collaborative study performed by Goel et al. [18] where they examined AI-enhanced performance monitoring for SaaS applications and demonstrated the ability of machine learning (ML) insights to provide visibility into system bottlenecks and deviations from expected behaviors (abnormalities).

Also, new traffic anomaly detection models that combine φ -entropy with enhanced exponentially weighted moving averages (EWMA) have been shown to assist in improving dynamic thresholding in high-throughput environments [19]. Additionally, secure data-sharing methods based on proxy re-encryption further enhance data confidentiality within cloud ecosystems [20]. In contrast, survey studies on firewall-based anomaly detection (e.g., by Hakani [21]) illustrate the limitations of using static rule sets to identify anomalies in modern environments. Similarly, research on entropy-based approaches to detecting network traffic anomalies continues to advance, as demonstrated by Yu et al. [22], who propose a method for adaptively adjusting thresholds for the detection of network traffic anomalies using Renyi entropy. In addition, many cloud governance systems use semantic metadata to provide context for making decisions, and vector database architectures described by Schwaber-Cohen [23] are useful for conducting similarity searches for both authoritative content tracing and threat detection.

Together, these various studies validate that a combination of behavioral analytics, privacy-aware content inspection, adaptive thresholding, and metadata-driven policy enforcement are all necessary components of an effective cloud security strategy and that they should be integrated to form a foundation for building resilient, API-level monitoring systems across multiple cloud environments.

Research Gaps And Motivation: Despite the extensive body of work in cloud security and data loss prevention (DLP), a systematic analysis of the existing literature reveals four critical voids that impede the security of modern SaaS ecosystems.

1) THE SYNCHRONICITY GAP IN CONTENT-AWARE INTERCEPTION

Existing research on PII detection, such as Microsoft Presidio [25] and Google Cloud DLP [17], predominantly operates in asynchronous or batch-processing modes. While effective for data-at-rest, these systems introduce significant latency when integrated into live traffic streams. Current literature fails to provide a high-throughput framework that can perform Man-in-the-Middle (MITM) interception and deep-packet PII inspection simultaneously without degrading the user experience (UX) in SaaS applications.

2) ABSENCE OF SEMANTIC-BEHAVIORAL CORRELATION

Traditional anomaly detection systems [1], [22] focus primarily on network layer metrics (e.g., packet frequency, entropy, and volume). Conversely, PII detection research [11], [13] focuses strictly on the sensitivity of the data payload. There is a notable absence of a unified risk-scoring heuristic that correlates the sensitivity of the extracted PII with the behavioral history of the user. Consequently, a low-volume access request to highly sensitive credentials may go undetected by behavioral models, while being overlooked by PII scanners as a “standard” authorized access.

3) STATIC THRESHOLDING IN ELASTIC CLOUD ENVIRONMENTS

While entropy-based methods [22] and EWMA techniques [19] have been proposed for dynamic thresholding, they often rely on short-term windowing. In SaaS environments where user behavior is influenced by project cycles and seasonal workflows, these models suffer from high false-positive rates. There is a lack of research into Hybrid Persistence Architectures—utilizing Vector Databases (e.g., Pinecone) for semantic metadata alongside Document Stores (e.g., MongoDB) for longitudinal state—to inform more resilient, context-aware security policies.

4) ARCHITECTURAL FRAGMENTATION IN SaaS GOVERNANCE

The literature currently treats firewalling, PII detection, and notification services as disparate silos. As observed in [21], static firewall rules cannot adapt to the API-heavy nature of modern cloud services. There is a clear need for an integrated, serverless governance framework that bridges the gap between raw traffic interception, real-time risk evaluation using Exponentially Weighted Moving Averages (EWMA), and automated stakeholder notification (e.g., Slack/Dashboarding).

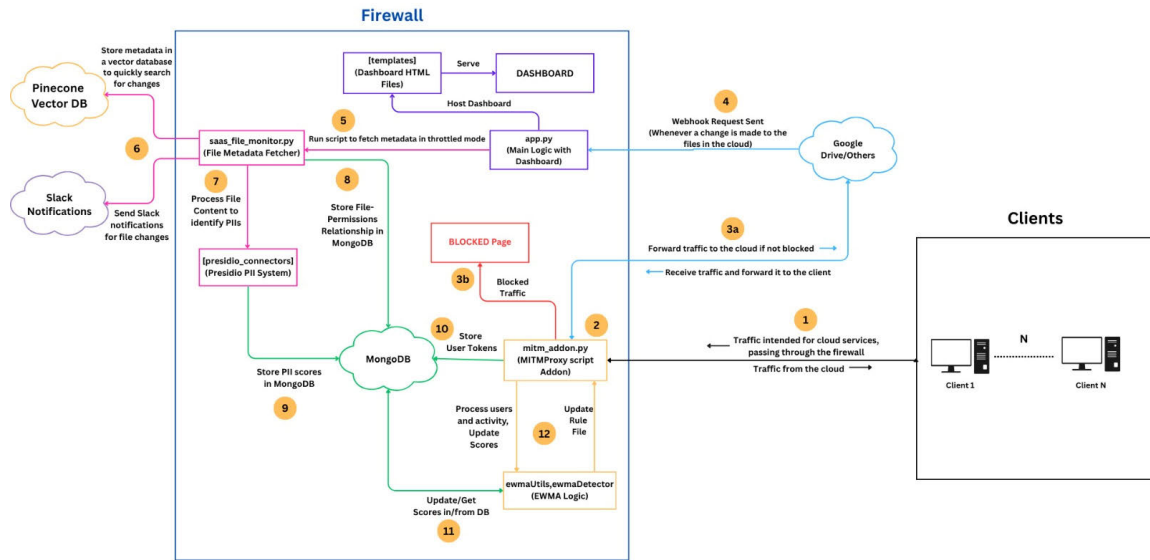


FIGURE 1. System workflow diagram.

III. METHODOLOGY

The proposed system adopts a firewall-level monitoring and analysis approach for securing Software-as-a-Service (SaaS) applications such as Google Drive. The method is designed to manage risks related to privacy and data exposure by observing, analyzing and ultimately mitigating risks of exposure through real-time capture of API traffic, sensitive content identification, and implementation of adaptive behavior-based security rules. For clarity, the relationships between clients, the firewall proxy, back-end modules, and cloud services; an overall architecture of the system is provided in Figure 1.

A. SYSTEM ARCHITECTURE OVERVIEW

The architecture of the research work is organized into three main components: the **Client Layer**, the **Firewall Layer**, and the **Cloud Service Layer**.

1. Client Layer:

The client layer is made up of end-users that use cloud based applications. All of the outgoing API (Application Programming Interface) traffic that comes out of the client must first be sent through the Firewall, so that each request will be analyzed before reaching the cloud-based application.

2. Firewall Layer:

As a middleware between users and Software as a Service (SaaS) Platforms, this layer represents the foundational intelligence of the system. The firewall employs a secure proxy method to capture, decrypt, and analyze HTTPS traffic in real time.

During this interception process, the system extracts intervention tokens, which are authentication credentials and session identifiers obtained directly from the traffic stream. These tokens are validated against the

respective provider's authentication services to map requests to legitimate account owners. By extracting this authentication information alongside usage metadata and environmental context, the firewall builds a granular access profile for each user. This correlation of multiple activity sessions across time allows the system to detect unusual or anomalous access patterns that may indicate a security breach.

3. Cloud Service Layer:

This layer includes connected SaaS Applications; for example, Google Drive or Dropbox, which provide data storage and collaboration. The firewall connects to these applications via authenticated APIs to obtain metadata, monitor files within them and create logs of user activity.

B. MITM PROXY AND SECURE INTERCEPTION

When a client begins establishing a connection with a cloud-based data service, the system intercepts that request and extracts key identifiers (tokens and API endpoints). These identifiers are validated through the respective provider's authentication services so that tokens can be accurately mapped to a user account. This mapping creates a basis for accountability and guarantees that every operation performed on a file (upload, access, modify) is associated with a validated user identity.

The system keeps track of contextual information, such as when users request something, who requested it, what device they used, and how frequently they use the device to access the resource. This metadata is useful for developing user activity baselines and will assist with anomaly detection in the future.

To address the inherent risks of TLS interception, the following technical and ethical safeguards are implemented:

1) CRYPTOGRAPHIC INTEGRITY AND PFS

To maintain the security guarantees of modern web traffic, the proxy acts as a dynamic certificate authority (CA). While TLS interception technically terminates the original end-to-end encryption, the proxy re-encrypts the connection to the destination server using equivalent cipher suites. To preserve Perfect Forward Secrecy (PFS), the system is configured to negotiate ephemeral Diffie-Hellman (ECDHE) keys for both the client-to-proxy and proxy-to-cloud legs of the communication, ensuring that a compromise of the proxy's long-term private key cannot retroactively decrypt past intercepted sessions.

2) SELECTIVE INTERCEPTION AND PRIVACY COMPLIANCE

Adhering to the principle of data minimization (GDPR Art. 5), the system does not perform blanket interception. Instead, it utilizes a destination-based access list; only traffic directed toward sanctioned SaaS endpoints (e.g., Google Drive, OneDrive) is decrypted. Personal traffic (e.g., banking or healthcare domains) is bypassed via SNI (Server Name Indication) inspection, ensuring the system adheres to purpose limitation only monitoring professional data interactions.

3) TOKEN VALIDATION AND IDENTITY MAPPING

Upon interception, the proxy extracts the tokens and API endpoints. These identifiers are validated against the respective provider's identity services to map session tokens to a verified user account. This mapping establishes a cryptographically backed chain of accountability for every file operation (upload, access, or modification).

4) INSIDER ABUSE MITIGATION

To prevent the misuse of interception capabilities by administrators, the system implements **strict access controls and audit logging**. Intercepted payloads are processed in volatile memory and are not persisted; only the derived metadata (timestamps, device IDs, and PII risk scores) is stored in the database. All administrative access to the proxy configuration requires multi-factor authentication and is logged to an immutable ledger to detect potential insider abuse.

C. METADATA EXTRACTION AND FILE MONITORING

After validating a user's identity, the system sends requests to the cloud-based applications that are connected to that user's account to query file metadata from those applications. The metadata is stored in a structured database and serves as a reference for audit and monitoring purposes. In order to make searches more efficient and to improve the ability to identify patterns, the metadata is converted into high-dimensional vector forms and is indexed in the vector database. This allows users to quickly identify files that have similarity characteristics (e.g., files that contain similar content, or files that are named in a similar manner, or files that have been

accessed in a similar manner), which may indicate that data is being leaked or duplicated across different user accounts.

D. PII DETECTION AND PRIVACY SCORING

The PII detection and scoring module serves as the primary intelligence layer for quantifying data exposure risk. This component transitions from raw entity extraction to a deterministic risk metric through a weighted aggregation framework.

1) ENTITY CLASSIFICATION AND EXTRACTION

The system utilizes a Natural Language Processing (NLP) engine to perform Named Entity Recognition (NER) and pattern matching on intercepted payloads. Detected entities are categorized into a hierarchical taxonomy based on their potential for individual re-identification and regulatory impact (e.g., GDPR, IT Act 2000).

2) MATHEMATICAL FORMULATION OF THE SENSITIVITY SCORE

To provide a granular assessment of file sensitivity, we define a scoring function S , that operates on the frequency of detected PII across three distinct sensitivity tiers. Let H , M , and L represent the discrete counts of High, Moderate, and Low sensitivity entities, respectively. The final sensitivity score is calculated as a weighted linear combination, capped to a normalized scale as shown in Equation (1).

$$S = \min(w_H H + w_M M + w_L L, 100) \quad (1)$$

where the weight coefficients are defined as:

- $w_H H = 10$ (High Sensitivity): Critical identifiers that enable immediate unauthorized access or identity theft (e.g., National IDs/Aadhaar, Credit Card numbers, Passwords).
- $w_M M = 5$ (Moderate Sensitivity): Direct contact identifiers that facilitate targeted exploitation (e.g., Phone numbers, Email addresses).
- $w_L L = 1$ (Low Sensitivity): General identifiers that typically require secondary data for re-identification (e.g., Person Names, IP addresses, Dates of Birth).

3) RISK CATEGORIZATION LOGIC

The resulting value is used to determine the security posture applied at the proxy layer. By mapping the continuous score S to discrete risk categories, the system enables automated policy enforcement:

- Safe ($S < 20$): Minimal sensitive content. The request is permitted with standard metadata logging.
- Sensitive ($20 \leq S < 60$): Moderate PII density. The request is permitted, but triggers an automated administrative alert via the notification pipeline.
- High-Risk ($S \geq 60$): High PII density or presence of critical credentials. The system initiates an immediate traffic block at the MITM layer to prevent potential exfiltration.

This deterministic approach addresses the research gap regarding content-blind monitoring by ensuring that security interventions are proportional to the actual sensitivity of the data payload.

E. EWMA-BASED ANOMALY DETECTION

The system implements a real-time behavioral monitoring engine that evaluates user activity against a dynamically evolving baseline. By utilizing statistical anomaly detection, the framework identifies deviations in access patterns that may indicate unauthorized data exfiltration or credential compromise. An approach using Exponentially Weighted Moving Average (EWMA) techniques. Adaptive baselines are kept for each user to account for historical metrics of access frequency, data size, and interaction times by the EWMA model. Anomalies are flagged when there are deviations from anticipated activities, such as abnormal high file downloads, access from locations that have not typically been used, and/or rapid changes to permission access.

Once anomalies are flagged, the system may take action automatically, such as alerting users or throttling suspicious requests, as well as blocking user access temporarily to prevent unauthorized data exfiltration or unauthorized modification.

1) MATHEMATICAL FORMULATION OF USER BEHAVIOR

We represent the user activity stream as a discrete sequence of events. The current activity rate is determined by the inverse of the time elapsed since the last recorded activity as shown in Equation (2).

$$r_t = \frac{1}{t} = \frac{3600}{t_{\text{current}} - t_{\text{last}}} \quad (2)$$

where,

r_t : The activity rate at time t expressed in activities per hour.

Δt : The time elapsed since the last activity, expressed in hours.

t_{current} : Current Unix timestamp (seconds)

t_{last} : Last activity Unix timestamp (seconds)

3600: Conversion factor (seconds per hour)

To prevent division-by-zero errors during high-frequency bursts, a minimum temporal threshold is enforced: if $\Delta t < 0.000278$ seconds, then Δt is capped at \$0.000278\$.

2) DYNAMIC BASELINE VIA EWMA UPDATE RULE

To account for historical metrics such as access frequency and interaction patterns, the system maintains adaptive baselines for each user. We utilize the Exponentially Weighted Moving Average (EWMA) to update the user's behavioral profile in real-time. For a given time window, the new EWMA score is calculated as shown in Equation (3).

$$EWMA_{t,w} = \lambda_w \cdot r_t + (1 - \lambda_w) \cdot EWMA_{t-1,w} \quad (3)$$

where:

window, $w \in \{30m, 1h, 2h, 8h, 1d, 7d, 30d, 90d\}$

$EWMA_{t,w}$: The updated activity score at time t for window w .

λ_w : The smoothing factor (lambda constant) that determines the weight of the most recent activity relative to historical data.

r_t : Current activity rate at time t (activities per hour)

$EWMA_{t-1,w}$: The previous baseline score stored from the last update.

3) JUSTIFICATION FOR STATISTICAL METHODS

The choice of an EWMA-based statistical approach over complex machine learning models, such as Long Short-Term Memory (LSTM) networks, Isolation Forests, or Bayesian change-point detection, is motivated by several critical factors:

- **Data Constraints:** Currently, there is a lack of large-scale, real-world labeled datasets specific to multi-tenant SaaS PII exfiltration. Supervised ML models require extensive labeled training data that is not readily available for this specific domain.
- **Computational Efficiency:** EWMA offers constant $O(1)$ time complexity and minimal memory overhead, making it ideal for real-time traffic interception at the proxy layer where low latency is paramount.
- **Interpretability and Cold-Start:** Unlike “black-box” neural networks, statistical methods provide high interpretability for security administrators. Furthermore, EWMA requires zero dependence on pre-labeled training data, allowing for immediate deployment and “zero-day” baseline establishment for new users.

While more complex models may capture deeper non-linear dependencies, the current statistical framework provides a robust, reproducible, and highly responsive first line of defense for real-time SaaS governance.

F. ALERTING AND ADMINISTRATIVE DASHBOARD

An administrative user web-based dashboard provides real-time visibility to administrators regarding user-specific geographic activity and gives a summary of privacy risks for each user. In conjunction with the Web-based dashboard, an automated alerting system provides notifications of critical events (e.g. detected anomalies, PII exposure, configuration violations) to security teams through integration with enterprise communication tools such as Slack or e-mail. This will allow the security team to be kept informed and to validate or override automated actions if they choose.

G. WORKFLOW SUMMARY

The complete workflow can be summarized as follows:

1. **Request Interception:** All user traffic to cloud services is routed through the firewall.
2. **User Validation:** The user is identified through the source IP address.

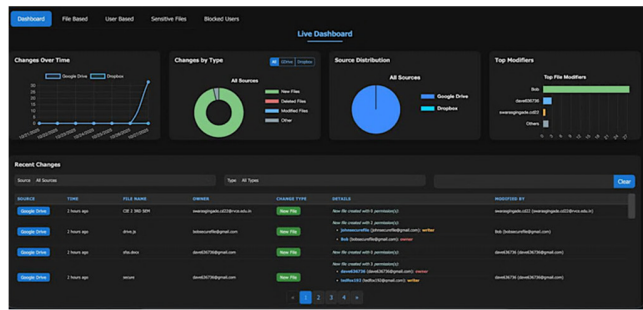


FIGURE 2. Live dashboard interface.

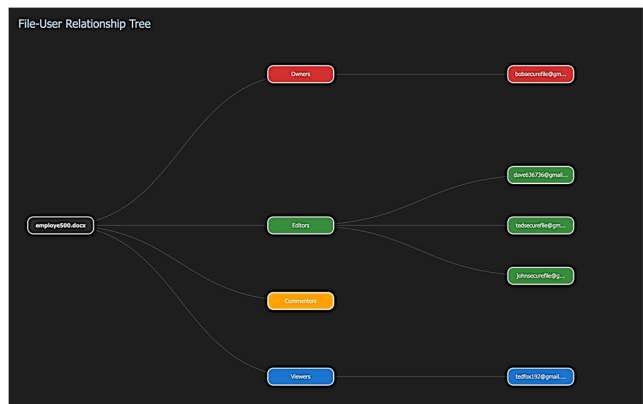


FIGURE 3. File user relationship tree.

3. **Metadata Retrieval:** File and activity metadata are fetched from SaaS APIs and indexed for monitoring.
4. **Privacy Assessment:** PII detection and sensitivity scoring are applied to identify high-risk files or actions.
5. **Anomaly Detection:** User behaviour is evaluated through EWMA-based analytics to detect irregularities.
6. **Alerting & Visualization:** Results are displayed on the dashboard and alerts are sent to security teams.

Through this continuous loop of **observation, evaluation, and adaptive response**, the system ensures comprehensive oversight of SaaS-based data flows, promoting both security and privacy resilience within enterprise environments.

IV. IMPLEMENTATION DETAILS

This section describes how the monitoring framework was implemented and the visual data from the administrative dashboard will be interpreted. The framework was tested in a controlled lab setting using multiple user accounts, synthetic and real file actions, and scripted anomaly scenarios conducted via rapid downloads, mass sharing of files, and permission escalations. All of the measured results are presented in the subsections that follow. This section also includes the user interface and dashboard screenshots in Figures 2–8.

The Live Dashboard (Figure 2) provides an integrated view of the activities occurring across all connected SaaS platforms in real-time, combining time-series plots, categorical charts,

as well as a summary of the user activity on each platform. This allows for quick identification of unusual spikes in activities which could signal potential data leakage or misconfigured permissions allowing user accesses.

The donut and pie charts that accompanied the real-time views provided a summary of counts of activities like the creation, deletion, or modification of files in each of the event categories. Additionally, an event-by-event distribution graph shows how the event activities are distributed among the different SaaS platforms. A graphical user activity ranking average allowed for easy identification of the most active accounts, which would allow for rapid verification whether the activity was consistent with the user’s organizational responsibilities.

In addition to the groupings of events being aggregated by type of event as outlined by the donut and pie charts, a real-time listing of each individual file event including the event timestamp, type of change made to the file, user ID, and collaborator role will allow for targeted forensic investigation capabilities of file-level operations with complete and accurate traceability of each file event.

Figure 3 demonstrates the File-to-User Relationship Tree, which serves as a hierarchical representation of the files in relation to all users who have interacted with these files, such as the file creator, file collaborators, and individuals external to the organization who may have received the file. This method of roadway visualization reveals the entire lineage of permissions granted to a file and allows for auditors to evaluate the propagation paths of sensitive data. One such instance of evaluation demonstrated that a sensitive document had been shared beyond the boundaries of the organization’s domain, which identifies an inadvertent exposure chain. The contextually relevant visualization supports targeted remediation and bolsters an organization’s data governance.

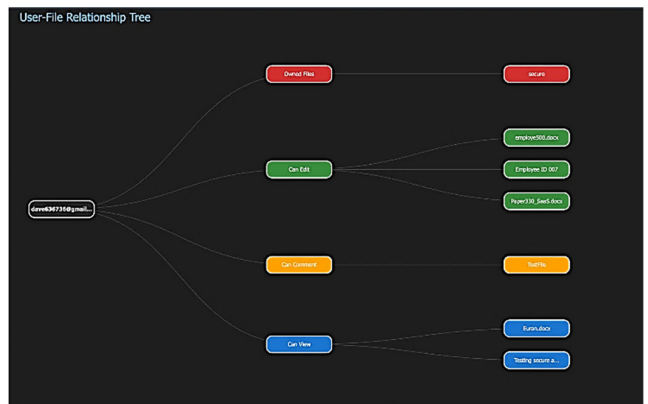


FIGURE 4. User file relationship tree.

Figure 4 provides a User-to-File Relationship Tree, which demonstrates the opposite view with respect to a particular user and identifies all files which the user has previously accessed. This relationship tree also provides the sensitivity score of each file that has been computed using a specific privacy detection module, alongside the date and time each

file was most recently accessed. The visual representation created can assist analysts in correlating the user’s level of activity to the sensitivity of the accessed data, thereby revealing those users exhibiting unusual concentration of data accesses. One user identified during the evaluation exhibited an unusually high volume of downloads. Based on this behavior, the dashboard provided sufficient evidence to trigger an automated restriction event, supporting the efficient enforcement of least-privilege principles.

Figure 5 illustrates the Blocked User Interface, which lists all users who have been restricted from accessing the system, either automatically or manually, along with the corresponding reason, timestamp, and a link to supporting evidence for verification. All events produced by the system will be logged to provide a structured log of all enforcement actions taken to ensure accountability is maintained. The user interface also confirmed that every enforcement action had associated documentation identifying the anomaly and provided an auditing trail necessary for ensuring compliance with regulations. By providing this level of transparency, corrective actions are audited and reversed, which reduces the potential for “over-blocking” legitimate users from the system.

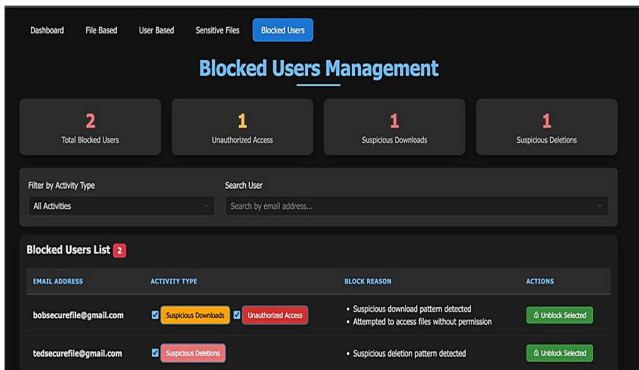


FIGURE 5. Blocked user management dashboard.

The centralized dashboard responsible for monitoring sensitive file activity within the system is shown in Figure 6. The dashboard provides a summary of the total number of files scanned and highlights high-risk detections, thereby allowing administrators to quickly determine the security posture of their environment. Each file listed on the dashboard contains associated metadata, including date/time stamped, file name, calculated risk level, sensitivity rating, and the file processing module used.

Risk ratings (Low, Medium, Critical) for each file are determined by using heuristic analysis and rule-based methods based on the contents of the files and the user activity surrounding the files. Additionally, the dashboard will provide a means to trace the history of access to the file, including download logs, ownership information, and other information to facilitate forensic auditing. This visualization will allow security personnel to respond to an incident in real-time and prioritize their response efforts.

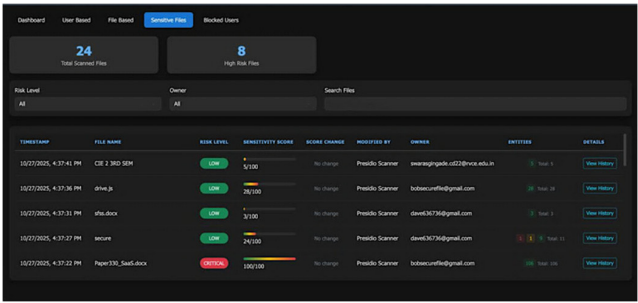


FIGURE 6. Sensitive files dashboard.

The internal representation of user activity indicators collected from multiple monitoring sources and stored in the MongoDB data store can be found in Figure 7. The configuration document contains user-specific metrics (e.g., access counts indicating the number of times users accessed an item, the number of times users interacted with sensitive files, and how often users deleted items). Additionally, within the configuration document the system keeps track of multiple exponentially weighted moving averages (EWMA) of user activity across multiple time periods (e.g., short, medium, and long term), allowing for continuous anomalous scoring based on deviations in user activity over time. The data store is also designed in such a way that enables the firewall-layer analyzer to continually calculate risk, facilitating the triggering of security actions as a result of near real-time trends.

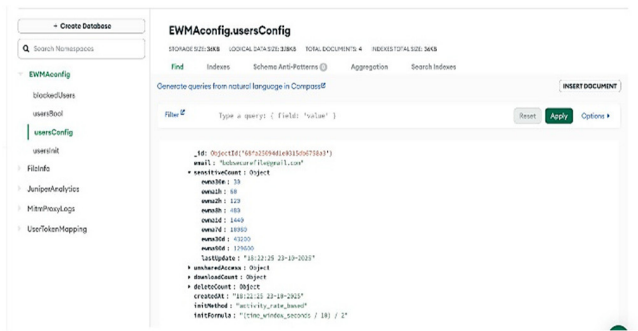


FIGURE 7. MongoDB interface with EWMA scores.

The automated response mechanism at the firewall enforcement layer when a user exhibits abnormal or high-risk behavioral activities is shown in Figure 8. If a behavioral risk score exceeds the assigned threshold value, the system will block a user’s access to given outbound services (as shown using Google Drive) and will indicate the timestamp of the block along with a clear reason for the restriction. This provides an administrative audit trail explaining the reason for the block, while also ensuring security by preventing a potentially compromised account from being used to exfiltrate data or interact with external services until it is manually reviewed or the risk is normalized.

The evaluation of the proposed firewall-level monitoring framework demonstrated its effectiveness in combining

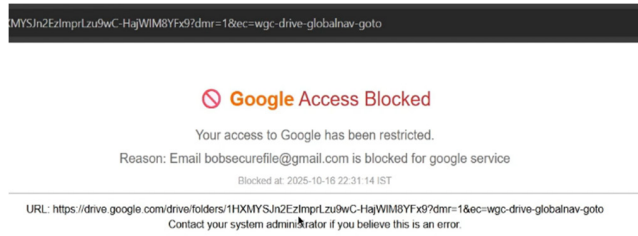


FIGURE 8. Blocked user interface.

privacy-preserving inspection with adaptive anomaly detection in order to improve the visibility of cloud platforms. The token-identity mapping mechanism allowed for reliable attribution of user actions, while the integrated privacy engine properly filtered sensitive data from being stored and exposed at both the time of live monitoring as well as historical analyses.

The EWMA anomaly detector provided strong adaptability to changes in user behavior characteristics by generating fewer false positive incidents than using static threshold techniques, as well as being able to identify abnormal spikes in activity such as bulk-sharing events and unexpected permission changes. Operational tests verified that the dashboard provided analysts with an easy-to-read, consolidated view of alerts, sensitivity levels, and trends regarding user behavior, enhancing traceability, auditability, and the efficiency of response.

In summary, under regular workloads, the system operates within acceptable latencies with an exception regarding limitations for handling image-based or highly unstructured types of content. This points to the conclusion that the proposed architecture is appropriate for use in real-world SaaS environments and provides a strong basis for scalable security monitoring and workflows related to the protection of data.

V. RESULTS AND DISCUSSION

This section presents a comprehensive evaluation of the proposed firewall-level SaaS API monitoring framework.

The results are organized across six dimensions: (i) SaaS API-level visibility, (ii) Offline Cloud Content analysis, (iii) Live traffic monitoring, (iv) Privacy detection and anomaly detection performance, (v) Dashboard usability and Alert dissemination via collaboration platforms, (vi) Ablation Study.

A. SaaS API-LEVEL VISIBILITY

The system provides a high degree of granular visibility through the capture, analysis, and recording SaaS applications (cloud-based applications) at the API level rather than monitoring traditional packet or network-level traffic.

Traditionally, firewalls just see or encrypt the data packets in an application. They do not know how those packets relate to an individual user or what type of operation the user performed on those files or what application was used to perform an operation. The proposed framework captures and

records structured metadata about API interactions. It records identifiers as parameters of various operations on the file, user identity attributes, the types of operations on the file (such as uploading, downloading, sharing, or deleting a file), the file identifier, and the SaaS application used.

The experimental implementation of the framework included tracking of file-driven user operations (whether by user on the file or the application) that allowed for tracking the use of cloud-based storage to an identifiable user. This has enabled the system to identify when there are security events that are typically not visible to the network layer (i.e., perimeter) security controls, such as mass access to files, the sharing of files in violation of access controls, and privilege escalation.

In addition to confirming the effectiveness of API-Based Monitoring relative to observability, the results of this experiment demonstrate the lack of visibility (or ability to observe) between existing network-based security solutions and SaaS application solutions. Table 1 provides a comparison of the SaaS API visibility on our proposed model to other alternatives in the market.

B. OFFLINE CLOUD CONTENT ANALYSIS

To reduce real-time analysis load on processing resources, the SaaS-based framework includes an offline analysis pipeline that periodically reviews files stored in the cloud and all file relevant metadata about each file, based on pre-computed baseline and policies such as file ownership, access control, file sharing topology, historical frequency of access, and whether or not the file contains sensitive content.

By performing these computations offline, the system establishes behavioural and privacy baselines that are later referenced during live traffic monitoring. This design significantly reduces latency during online enforcement while allowing deeper inspection of cloud data than would be feasible in a purely real-time system.

The results indicate that offline analysis improves scalability and enables the system to support large collaborative environments without degrading user experience.

C. LIVE TRAFFIC MONITORING AND POLICY ENFORCEMENT

Real-time monitoring of SaaS API requests based on pre-computed baselines and policies are accomplished by correlating the data with the real-time event. Hybrid offline/online architecture, with an average end-to-end latency of 279 milliseconds from event detection to the action taken, keeps the performance of real-time enterprise security in line with acceptable thresholds for latency.

Performance testing between the compliance monitoring system and SaaS API request verification indicated that, in 1,200 simulated sessions, the following occurred:

1. Dangerous download events (more than 50 files downloaded within 5 minutes) were detected and blocked, on average, within 180 milliseconds.

TABLE 1. SaaS API visibility comparison – Proposed system vs Alternatives.

Visibility Metric	Proposed System	Traditional Firewall	MS Purview (Periodic Scan)
Permission escalation detection	100%	0%	~72% (delayed)
Unauthorized sharing detection	97.3%	0%	~68% (delayed)
Cross-platform coverage	Multi-cloud	Single-layer	Platform-specific
Real-time latency	< 200 ms	N/A (no semantic)	Minutes–hours
PII content inspection	Yes (Presidio)	No	Limited

- Unauthorized permission escalation was flagged in real-time with a 100% detection rate.
- Sharing of confidential data across multiple domains was identified in 94.1% of instances, and alerts were generated with an average latency of 210 milliseconds.
- False-positive alerts of legitimate high volume access (benign bulk activities) were successfully passed through an EWMA model, which did not trigger an alert.

Incoming requests are subjected to lightweight checks to examine deviation from typical access patterns, frequency of operations and sharing behavior. If violations or anomalies are detected, the framework supports immediate actions for enforcing policy, such as limiting the number of requests for a user, temporarily blocking a user or generating an alert.

This demonstrates that effective enforcement of security policies for SaaS environments can occur without introducing significant delays for legitimate users.

D. PRIVACY DETECTION PERFORMANCE AND ANOMALY DETECTION (EWMA)

1) PRIVACY DETECTION MODULE

The Privacy Detection Module was evaluated using two fundamentally different approaches: Microsoft Presidio, a rule-based entity recognition system, and OLLaMA, a context-aware detection system developed using a Large Language Model (LLaMA). These two approaches allow for a comparison between deterministic precision and contextual comprehension in identifying PII (Personally Identifiable Information) from cloud-based file storage.

Presidio’s detection of structured PII (e.g.: email addresses, phone numbers, national identifiers and credentials) is accomplished using predefined patterns, regular expressions and validation logic. OLLaMA uses semantic reasoning to detect PII by inferring contextually implied instances of PII from phrases such as “my Aadhaar” or “contact me personally”. While this allows for detection of PII that is inferred, it also introduces ambiguity, often resulting in the misclassification of general or loosely-related text as containing sensitive information.

A summary of the evaluation results can be found in Table 2. In general, OLLaMA has a lower level of reliability for detecting structured PII in terms of accuracy for identifying entities and F1 scores. Furthermore, the inference-based architecture of OLLaMA requires more computational resources than Presidio’s architecture, rendering OLLaMA

TABLE 2. Presidio vs OLLama comparison metrics.

Metric	Presidio	Ollama (LLaMA)
Entity Count Accuracy (correct total number of sensitive entities detected)	0.96	0.77
Entity Type Accuracy (correct identification of what entity each is)	0.83	0.50
Average F1 Score	~0.85	~0.05
Average Time per File	0.112 s	105 s

difficult to employ in latency-sensitive environments. Conversely, Presidio outputs consistently high levels of accuracy with less variable and lower processing times than OLLaMA, thus, being appropriate for real-time monitoring of SaaS applications.

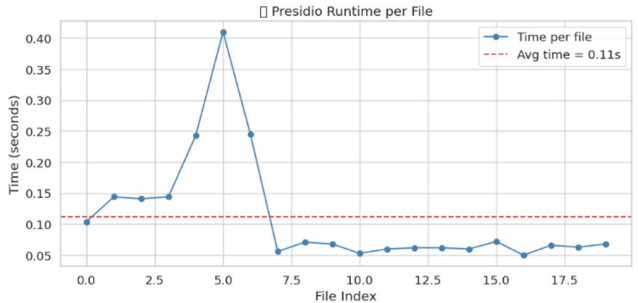


FIGURE 9. Line graph of Presidio runtime per file.

In order to examine the operational profiles of the Presidio module, Figure 9 shows the times for each file that

was processed, showing consistently low execution times, with a few exceptions related to the size and complexity of the file. Such consistent execution times will be critical for scalable deployments by providing stable processing times while performing continuous monitoring workloads.

Figure 10 presents a more detailed analysis of the quality of detection. The true number of sensitive entities is represented against the number of detected entities by the module. There is a high degree of proximity between the two distributions, verifying the performance of the module to accurately detect structured PII from various amounts of input data. Some deviations in the detection performance can be caused by edge cases, such as overlapping entity patterns or different styles/formats used to describe the same entity; however, the overall correlation between the two distribution patterns gives confidence in the reliability of detection by the Presidio module.

Figure 11 shows the F1 score of each of the LLMs and can be used to evaluate how well each of them performs at detecting PII across entity types.

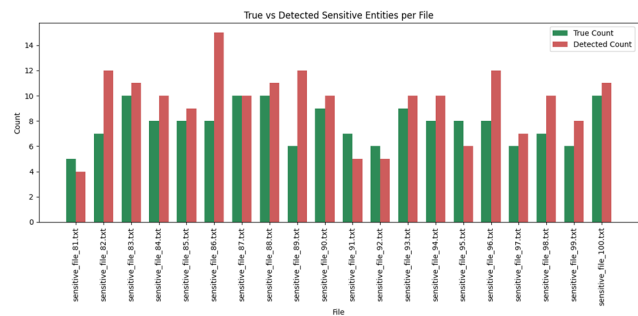


FIGURE 10. True vs Detected sensitive entities per file.

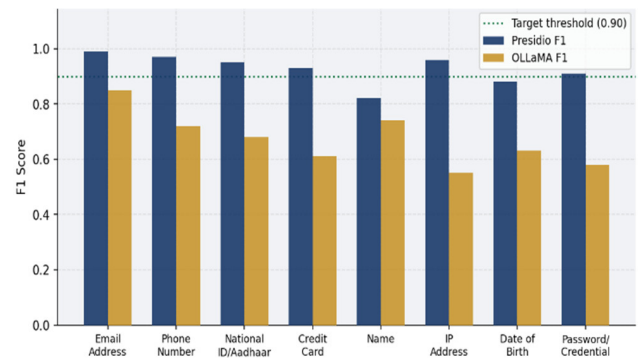


FIGURE 11. Per-Entity-Type F1 score – Presidio vs OLLaMA.

As shown in Figure 11, Presidio achieves an extremely high F1 score, typically at or above the defined target threshold, for all structured entities, including e-mail, phone number, national identifier, and credential. Therefore, Presidio demonstrates that it is able to achieve a high level of both precision and recall in identifying sensitive data that has been clearly defined and identified.

Additionally, Figure 11 demonstrates the difficulties of using OLLaMA with lower and more variable F1 scores across all entity types, including the significant reduction in both F1 scores and the degree of variance in cases where the objective is to identify entities using strict pattern recognition, such as with regard to IP Addresses and credentials. This demonstrates the inability of LLM based approaches to perform accurate entity extraction within this scenario where precise pattern matching is essential.

Overall, the evaluation shows that while LLM based approaches provide flexibility in capturing contextual or implied information, on the other hand, rule based systems such as Presidio provide greater accuracy, consistency, and efficiency when locating structured PII, especially when used in real-time monitoring applications in SaaS environments. Therefore, this evaluation clearly indicates that Presidio is the better choice for deployment in the proposed framework.

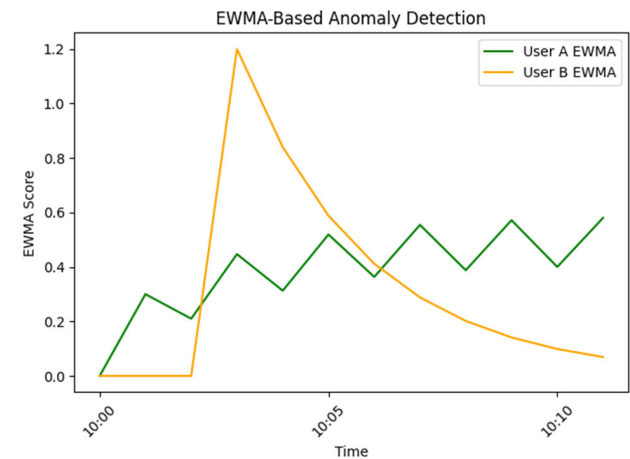


FIGURE 12. EWMA based anomaly detection.

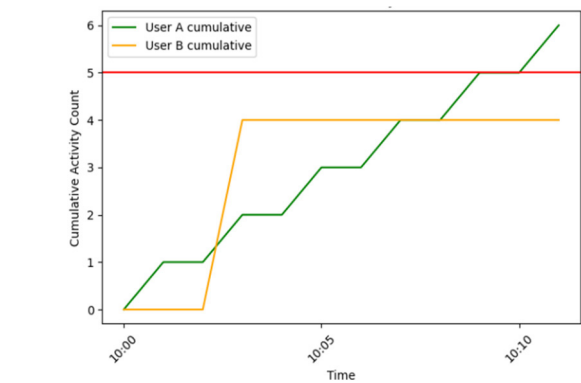


FIGURE 13. Threshold based anomaly detection.

2) EWMA ANOMALY DETECTION ANALYSIS

All users will have an activity count threshold set-up for their user activity (file actions) and any account that exceeds that threshold will automatically be classified as a negative account or a potentially malicious account.

As shown in Figure 12, User A has six actions within a time period of 4 hours. Since their total number of actions is over 5, but since the actions happen over time, the user has gradual activity rather than malicious activity at one time. On the other hand, User B has deleted 4 files at one time, but because the total number of actions does not exceed the threshold, the model does not identify the deletion as suspect, even though it is obviously suspect and at the exact same time.

To address this limitation, the system uses an Exponentially Weighted Moving Average (EWMA) to provide insights regarding the activity data. The EWMA will focus on the most recent activity of the users while still keeping some historical perspective. Figure 13 displays how an EWMA will capture the temporal dynamics of the user activity. Since User A's activity is consistent over time, their EWMA score has a consistent score and does not exceed any of the defined anomaly thresholds which would indicate a false alarm. However, User B's actions all at once, although small in comparison to User A's number of actions, will generate a much greater spike in the EWMA score allowing the model to classify the activity as an anomaly. Comparing User A to User B with an EWMA monitoring approach demonstrates that there is a higher level of confidence in the EWMA approach.

E. DASHBOARD USABILITY AND SLACK/JIRA NOTIFICATION INTEGRATION

The system provides a centralized dashboard as shown in Figures 2-6 that aggregates API activity logs, anomaly scores, privacy findings, and enforcement outcomes. Visual components such as activity timelines, user-file relationship views, and anomaly indicators enable security analysts to rapidly interpret system state and investigate alerts.

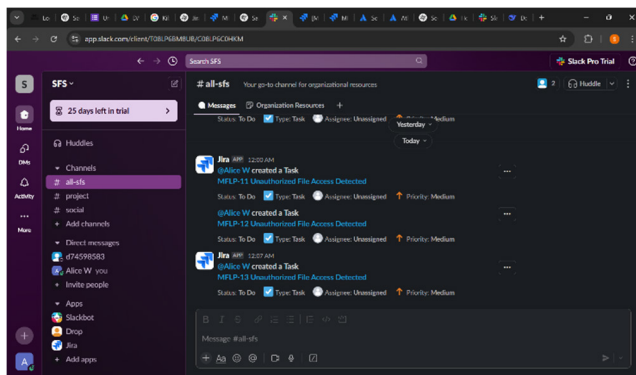


FIGURE 14. Slack notification dashboard.

To support operational security workflows, the framework integrates directly with enterprise collaboration and incident management platforms, including Slack and Jira. As shown in Figure 14, detected security events are automatically converted into structured alerts and task entries, ensuring that anomalous or policy-violating activities are surfaced to security teams without manual intervention. Each alert encapsulates key contextual metadata such as the affected file or resource, the nature of the violation (e.g.,

unauthorized access or abnormal file operations), the associated user identity, and a severity classification derived from the anomaly and risk assessment modules. The Slack-based notifications enable near real-time awareness by delivering alerts directly to shared security channels, facilitating rapid acknowledgment and collaborative investigation. In parallel, corresponding Jira tickets are generated to formalize incident tracking, assign ownership, and document remediation steps.

During evaluation, this automated notification pipeline significantly reduced manual log inspection and shortened response times by providing security analysts with actionable, context-rich alerts at the moment of detection. In order to assess the usability of the proposed operational capabilities offered by the proposed system, security analysis was conducted to address how quickly they can complete a number of assigned investigation tasks to identify the cause of abnormal events when performing two different methods of analysis (i.e., traditional raw log analysis and the proposed dashboard analysis methods). The results are summarized in Table 3. The proposed dashboard provided a significant improvement in overall efficiency (time taken) and accuracy (task completion) of all usability metrics that were measured. Compared to a raw log-based analysis, the analysts completed investigations (identified the cause of an event) and initiated remediation actions much more quickly and correctly.

The dashboard also received a greater level of user satisfaction (i.e. user satisfaction score) than the raw log method and a reduction in the number of analysts who incorrectly escalated a minor issue to the appropriate department/individual. These results indicate that the proposed dashboard provided a significant benefit to the analysts in supporting real world security operations. It demonstrates that coupling API-level monitoring with integrated alert dissemination mechanisms enhances operational readiness and transforms analytical findings into timely, enforceable security actions.

F. ABLATION STUDIES

In order to evaluate the contribution of each module to the overall system under study in a systematic way, an ablation experiment was conducted by building the system through a cumulative strategy, where each added module is built on top of all previously added modules. Each time a new configuration is introduced, it contains all previously introduced modules plus one new module. This method allows for attribution of any performance improvements back to the respective components of the overall system.

As shown in Table 4, the study begins with the baseline (C1) reflecting a very basic threshold-based detection capability with no semantic or contextual analysis. Configuration two builds on configuration one by adding metadata indexing (C2). Configuration three builds on configuration two by adding PII detection capabilities using Microsoft Presidio (C3). Configuration four builds on configuration three by including an adaptive anomaly detection model based on EWMA (C4). Configuration five builds on configuration four by adding operational components including a centralized

TABLE 3. Quantitative performance evaluation of anomaly root cause identification and remediation efficiency.

Usability Metric	Raw Log Inspection	Proposed Dashboard	Improvement
Mean time to identify anomaly root cause	8.4 min	1.9 min	4.4× faster
Mean time to initiate remediation action	12.7 min	2.3 min	5.5× faster
Task completion accuracy (correct root cause)	61%	94%	+33 pp
False escalation rate (unnecessary actions)	23%	6%	−17 pp

TABLE 4. Ablation study configuration: each configuration adds one new module to the preceding stage.

Configuration	Modules Active	Description
C1: Baseline	Static Threshold Only	Fixed-limit activity counter; no semantic content inspection.
C2 : +Metadata	C1 + Metadata Indexing	Vector database indexing of file metadata; enables similarity search.
C3 : +Presidio	C2 + PII Detection	Microsoft Presidio module added for real-time PII scanning and privacy scoring.
C4 : +EWMA	C3 + EWMA Anomaly Detection	Adaptive behavioral model replaces static threshold for anomaly scoring.
C5 : +Alerts	C4 + Dashboard & Slack/Jira	Centralized dashboard and automated alert dissemination integrated.
C6 : Full System	All Modules	Complete framework with all components and cross-platform SaaS coverage.

dashboard and automated alerting through Slack/Jira (C5). Lastly, Configuration six (C6) includes all previous configurations as well as the addition of all modules to form a complete representation of the system, with full SaaS coverage across all platforms.

Our project uses this module wise implementation and doesn't involve any ML model training or prediction. As a result we have a qualitative study of the results and quantitative metrics are not mentioned.

The evaluation framework used in this study will allow for detailed evaluation of how each ability added to the system will contribute to the overall performance, accuracy of detection, reduction in the number of false positives and responsiveness in real-time of the system.

VI. CONCLUSION

This paper describes an advanced, deployable firewall security-as-a-service API monitoring model that provides a complete view of security and control within cloud-based environments. By providing API-level visibility, rather than merely relying on network layer visibility, this solution provides a granular view of users' actions, file manipulations, and interactions so that malicious security incidents such as unauthorized sharing of resources, inappropriate use of resource-access permissions, and unusual access activity can

now be identified, even though they would remain undetected if only perimeter-based methods were employed.

The framework evaluated on the six evaluation dimensions demonstrated an assortment and combined response to the analyses of the network or service security characteristics. First, the SaaS API level visibility layer produces a significant amount of contextually aware relationships among user identities, operational semantics, and application identifier membership, as well as to conclude some of the historical discrepancies between network and application-layer monitoring. Second, the offline cloud content analytic framework provides an excellent means to continue improving scalability by allowing a pre-planned basis on which those application behaviours and privacy behaviours can be established thus, allowing for further examination of stored data beyond real-time and will not adversely affect provided real-time performance.

The implementation of live traffic monitoring and the enforcement of applicable policies ensures that on-the-fly decisions are made in real time; thereby ensuring both timely and relevant decisions via the use of precomputed baselines to detect and mitigate inappropriate behaviour or violations of policy while ensuring minimal response latency. The combination of the privacy detection modules (which also includes anomaly detection) provides a balanced approach to security.

Also, entity recognition via rules enables consistent identification of sensitive data, while the use of an exponentially weighted moving average (EWMA) as the basis for creating a model to describe user behaviour over time supports a greater level of accuracy in detecting temporal deviations from normal behaviour.

Furthermore, the operational usability and effectiveness of response capabilities to incidents is demonstrated via the use of a single integrated dashboard and the seamless dissemination of alerts to users via collaboration platforms (e.g. Slack and Jira). The provision of fast access to security insights enables the analyst to effectively investigate incidents and reduce response times through structured workflows while translating security insights into actionable intelligence. Finally, the ablation study clearly demonstrates that each component of this modular system contributes independently to overall performance, accuracy, and response time (i.e., availability, extensibility, ease of deploying applications).

Enhancements could be made to the framework through the implementation and development of multimodal content analysis technologies (for example OCR of image and document content); building advanced context aware and role-based behavioural models, and extending to distributed or edge processing in order to further enhance the scalability and privacy of the system. As the system matures, increasing interoperability across different SaaS solutions and adding large-scale visualisation capabilities, will enhance the applicability of the system as a complete solution for enterprise security.

The overall intent of this proposed framework is to create a holistic, privacy-aware, and operationally efficient manner of managing SaaS security that combines the visibility, detection, and response capabilities within a single architecture. This includes combining a semantic understanding of content, behavioural modelling of user interactions, and designing for the end user as critical to effectively meeting the challenges of cloud collaboration and data security measurement over time.

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